RouteRover

Project Team

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Chapter 1

Introduction

This document outlines the objectives, scope, and features of the RouteRover app. RouteRover is a location-based app that provides real-time information, navigation, safety alerts, and AI-driven recommendations. It differs from existing map systems by offering unique features like volunteer support, crowdsourced crime detection, and AR for interactive navigation. RouteRover is designed to enhance travel experiences by providing personalized recommendations, real-time updates, and multilingual support. This document serves as a guide for stakeholders and developers to ensure the app's successful implementation. detailed background of the system. [1].

1.1 Problem Statement

Existing navigation apps often fail to provide users with real-time, personalized safety information. While they offer directions and traffic updates, they lack dynamic features like crime alerts, local guidance, and interactive navigation. As a result, travelers frequently face challenges such as unfamiliar environments, safety concerns, and limited access to reliable, real-time data. RouteRover aims to address these shortcomings by incorporating crowdsourced safety updates, volunteer support, and AI-driven recommendations. This innovative approach will provide users with a safer and more interactive travel experience, making their journeys more enjoyable and efficient.

1.2 Scope

The scope of RouteRover is to develop a mobile application that will provide users with real-time, community-driven insights for safer more effective navigation. The main goal of this app is to provide real-time information on traffic, events, and crime while utilizing volunteer assistance and crowdsourced data. Users can customize notifications according to their tastes and preferences, so they will always be aware of important events like protests or red zone warnings in their area. Tourists, people who commute daily, and local communities looking for up-to-date information on their surroundings are the target audience for this application. The app will also support other languages, making it accessible to a worldwide user base. Additionally, RouteRover will provide local volunteer management, allowing customers to communicate with representatives in their region for chat and phone support. The scope includes developing essential modules to guarantee correct and dependable data, including, live updates, crime-detection and user generated content validation. In order to improve user engagementm, the app will also include futuristic features like gamification and predictive analytics. This project's limit are essentially determined by its capacity to deliver customized, real-time updates via a safe and scalable platform. The system will primarily concentrate on mobile solutions with the possibility of adding web-based interfaces in future.

1.3 Modules

The proposed RouteRover project consists of the following main modules. These are grouped based on the types of systems involved, such as the Client Mobile App and the Admin Web App.

1.3.1 Module 1

: **Client Mobile App** [Delivering the main user experience, supplying real-time data, and facilitating volunteer communication are the main objectives of this module. Users can receive timely alerts for safer navigation and customize the app's settings.]

1. Feature 1 : . Real-time Updates: Depending on the user's location, provide real-time traffic, crime, and event updates.

2. Feature 2 : Volunteer Interaction: Enable users to request help from nearby volunteers via phone or chat.

3. Feature 3: Custom Alerts: Allow users to personalize alerts according to their preferences, including warnings for crime or protest areas and safety issues.

4. Feature 4: Crowdsourced Data: Users provide location-based information, which is validated and disseminated to others. Examples of this information include traffic patterns and possible risks

1.3.2 Module 2

: **Admin Web App** The Admin Web App module is responsible for managing the application's data and ensuring that the crowdsourced information is true and dependable. In addition to overseeing volunteers and system users, administrators can verify and control user-generated content

1. Feature 1 : Data Validation: Verify the accuracy of all user-generated data, including traffic updates and criminal alerts, before sharing it with app users (a process known as "data validation").

2. Feature 2 : Volunteer Management: Keep an eye on and delegate work to neighborhood volunteers who communicate with users via the app.

3. Feature 3 : System Security: Take care of user access controls and maintain the privacy of the app's data.

1.4 User Classes and Characteristics

User Class	**Description**
User	A General User is an individual who uses the RouteRover app for traveling, checking real-time traffic, crime, and event information, and setting up personalized notifications. They can also share incidents or contribute crowdsourced information about their environment. Potential users come from a mix of daily commuters, tourists, and locals across different areas. Approximately 70% of users are predicted to use RouteRover regularly for safer guidance, with the most usage during morning and evening rush hours.
Volunteer	Volunteers are people who provide help to individuals in particular domains.They might answer questions or assist with requests via chat or phone. Vol-unteers will receive specialized training on utilizing the RouteRover app and engaging with users. Approximately 30% of volunteers are regularly en- gaged each day, and they might serve in regions that experience high levels of user requests, like busy urban areas or places plagued by traffic problems.
Administrator	The Administrator is responsible for overseeing the system, which includes validating user-generated data, supervising volunteers, and ensuring overall system security. Administrators are accountable for authenticating the correctness of crowdsourced data and will be granted entry to the admin web app. They are responsible for overseeing user accounts, maintaining system performance, and addressing security concerns. Administrators must undergo training in managing sensitive data and preserving the platform's integrity.
Local Guide	Local Guides are individuals with extensive knowledge of particular areas who help tourists and new residents by providing directions, suggestions, and information about the area. They provide more customized assistance than regular volunteers and can be called upon for specific questions. Guides will communicate using both chat and voice features within the app. During peak tourist seasons, around 20% of RouteRover users might seek help from local guides.

Table 1.1: User Classes and Characteristics for RouteRover

Chapter 2

Project Requirements

This chapter describes the functional and non-functional requirements of the project.

2.1 Use-case/Event Response Table/Storyboarding

Use Case Diagram:

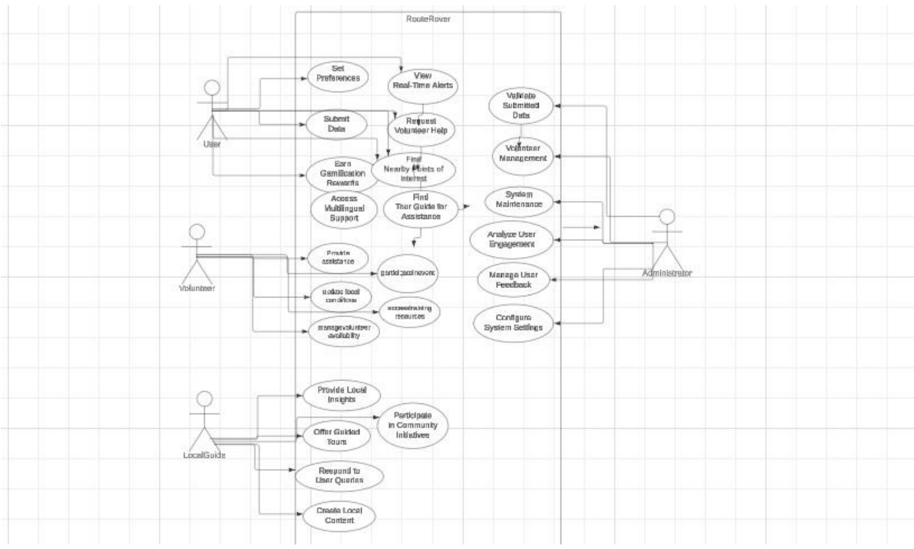


Figure 2.1: Use Case Diagram for RouteRover.

Event ID	Description(Action)	Trigger	System Response	User Feedback
1	User opens the app	User launches the RouteRover app	The app loads the map interface displaying the user's current location and relevant alerts.	User sees the map and gets an overview.
2	User sets alert preferences	User selects alert types from the settings menu	System saves user preferences and adjusts notification settings accordingly.	Confirmation message displayed.
3	New real-time alert generated	System detects a new traffic incident or crime	The app sends a push notification and updates the map interface with the new alert.	User receives a notification on their device.
4	User submits data	User reports local conditions (e.g., traffic jam)	System validates the submission and forwards it for admin review before publishing.	Confirmation that submission is in progress.
5	User requests volunteer help	User clicks the "Request Help" button	The app connects the user to a nearby volunteer via chat or voice call, displaying volunteer info.	User is connected with a volunteer.
6	Admin validates user-submitted data	Admin reviews user-submitted reports	The system updates the status of the submission and publishes validated data for all users.	Users see the updated information in the app.
7	User earns rewards	User's submitted data is validated	System credits points/badges to the user's account based on their contributions.	Notification of earned rewards appears.
8	User chooses a preferred language	User selects a language option in settings	The app changes the language of the interface and any voice interactions to the selected language.	All text and audio adjust to the chosen language.
9	Volunteer updates local conditions	Volunteer inputs new data about conditions	System validates and publishes the updated information to all users.	Users receive updates on the changes.
10	User views a historical event report	User selects the historical events option	The app retrieves and displays past alerts and conditions for the selected area.	User accesses historical data for insights.

Figure 2.2: Event Response Table for RouteRover.

2.2 High-Level Use Cases

2.2.1 Set Preferences

- **Use Case:** Set Preferences
  Actor: User
  Type: Primary
  Description: Customers have the ability to personalize their notification preferences, such as selecting their preferred categories for alerts like traffic, events, and crime.

2.2.2 View Real-Time Alerts

- **Use Case:** View Real-Time Alerts
  Actor: User
  Type: Primary
  Description: Individuals have access to instant alerts on traffic conditions, local events, and safety warnings in their vicinity.

2.2.3 Submit Data

- **Use Case:** Submit Data
  Actor: User
  Type: Primary
  Description: Individuals have the ability to inform others about nearby circumstances, activities, or concerns regarding safety, adding to the collective database formed by the community.

2.2.4 Request Volunteer Help

- **Use Case:** Request Volunteer Help
  Actor: User
  Type: Primary
  Description: Users are able to ask for help from volunteers in the area for advice or information about their environment.

2.2.5 Earn Gamification Rewards

- **Use Case:** Earn Gamification Rewards
  Actor: User
  Type: Primary
  Description: Users are able to participate in app activities to receive rewards, points, or badges, which promote consistent engagement and involvement.

2.2.6 Access Multilingual Support

- **Use Case:** Access Multilingual Support
  Actor: User
  Type: Primary

Description: Users have the option to choose their desired language for app interactions and notifications, improving accessibility.

2.2.7 View Local Business Recommendations

- **Use Case:** View Local Business Recommendations
  Actor: User
  Type: Primary
  Description: Users can receive suggestions for nearby businesses, eateries, and services tailored to their likes and current location.

2.2.8 Find Nearby Points of Interest

- **Use Case:** Find Nearby Points of Interest
  Actor: User
  Type: Primary
  Description: Users are able to explore nearby points of interest, offerings, or activities according to their personal preferences and current data.

2.2.9 Provide Assistance

- **Use Case:** Provide Assistance
  Actor: Volunteer
  Type: Primary
  Description: Volunteers are able to assist users by responding to questions or giving immediate information.

2.2.10 Update Local Conditions

- **Use Case:** Update Local Conditions
  Actor: Volunteer
  Type: Primary
  Description: Volunteers have the ability to ensure the accuracy and reliability of the data by updating the app with current local conditions, events, or safety issues.

2.2.11 Manage Volunteer Availability

- **Use Case:** Manage Volunteer Availability
  Actor: Volunteer
  Type: Primary
  Description: Volunteers have the option to specify when they are available to help users or answer requests.

2.2.12 Participate in Community Events

- **Use Case:** Participate in Community Events
  Actor: Volunteer
  Type: Primary
  Description: Volunteers have the option to either help plan or take part in nearby gatherings that focus on safety and community involvement.

2.2.13 Access Training Resources

- **Use Case:** Access Training Resources
  Actor: Volunteer
  Type: Primary
  Description: Volunteers can utilize training materials or resources to improve their ability to assist users effectively.

2.2.14 Provide Local Insights

- **Use Case:** Provide Local Insights
  Actor: Local Guide
  Type: Primary
  Description: Local experts offer insights on hidden gems, attractions, cultural facts, and helpful tips to improve visitor experiences.

2.2.15 Offer Guided Tours

- **Use Case:** Offer Guided Tours
  Actor: Local Guide
  Type: Primary
  Description: Local guides have the ability to design, plan, and oversee guided tours

that users can reserve, providing a organized discovery of the area with individualized knowledge.

2.2.16 Find Tour Guides for Assistance

- **Use Case:** Find Tour Guides for Assistance
  Actor: User
  Type: Primary
  Description: Users have the ability to look for and get in touch with nearby guides who can join them on their adventures, offering immediate help and knowledge.

2.2.17 Participate in Community Initiatives

- **Use Case:** Participate in Community Initiatives
  Actor: Local Guide
  Type: Primary
  Description: Local guides play an active role in local events and community projects, advocating for safety, awareness, and community cohesion.

2.2.18 Respond to User Queries

- **Use Case:** Respond to User Queries
  Actor: Local Guide
  Type: Primary
  Description: Local guides offer personalized help by answering users' inquiries on local sights, safety advice, and cultural customs.

2.2.19 Create Local Content

- **Use Case:** Create Local Content
  Actor: Local Guide
  Type: Primary
  Description: Local guides have the opportunity to share articles, blogs, or tips on the app, enhancing the content for users and showcasing local culture and events.

2.2.20 Validate Submitted Data

- **Use Case:** Validate Submitted Data
  Actor: Admin
  Type: Primary
  Description: Administrators have the ability to inspect and approve data provided by users to ensure it is accurate and reliable.

2.2.21 Volunteer Management

- **Use Case:** Volunteer Management
  Actor: Admin
  Type: Primary
  Description: Administrators have the ability to oversee volunteer profiles, track their activity, and facilitate volunteer initiatives. .

2.2.22 System Maintenance

- **Use Case:** System Maintenance
  Actor: Admin
  Type: Primary
  Description: Administrators have the ability to carry out maintenance activities in order to guarantee the system's dependability and resolve any technical problems.

2.2.23 Analyze User Engagement

- **Use Case:** Analyze User Engagement
  Actor: Admin
  Type: Primary
  Description: Admins have the ability to assess user engagement data in order to pinpoint patterns and areas that need enhancement..

2.2.24 Manage User Feedback

- **Use Case:** Manage User Feedback
  Actor: Admin
  Type: Primary

Description: Administrators have the ability to assess, reply to, and put into effect alterations according to user suggestions in order to improve app performance.

2.2.25 Configure System Settings

- **Use Case:** Configure System Settings
  Actor: Admin
  Type: Primary
  Description: Administrators have the ability to set up system settings, such as user roles, types of notifications, and policies for managing data.

2.3 Functional Requirements

2.3.1 Module 1

: **Client Mobile App** Following are the requirements for module 1:

1. FR1: Deliver immediate updates on traffic, crime, and events tailored to the user's location, while enabling users to communicate with local volunteers for help via phone or chat.

2. FR2 : Allow users to tailor alerts based on their preferences (such as crime or safety alerts) and submit verified crowdsourced data, while backing individualized notifications and user-friendly customization of app preferences.

2.3.2 Module 2

: **Admin Web App** Following are the requirements for module 2:

1. FR1 : Verify information created by user like traffic and crime updates prior to distributing to app users, while overseeing and controlling community volunteers engaging with users.

2. FR2 : Ensure the security and privacy of the system and user data, and facilitate efficient management of user accounts and access controls.

2.4 Non-Functional Requirements

This section specifies nonfunctional requirements. These quality requirements should be specific, quantitative, and verifiable. The following are some examples of documenting guidelines.

2.4.1 Reliability

The RouteRover app must remain highly reliable, with rare occurrences of system failure. The Mean Time Between Failures needs to exceed 1000 hours. A failure occurs when the application is not available for user interaction for longer than 5 minutes. In orderto reduce failures, redundant servers and real-time monitoring will be implemented by the system. Automated logging and user feedback will be used for error detection, with prompt updates and maintenance for corrections.

2.4.2 Usability

Emphasis is placed on creating an interface that is easy for users to navigate in usability requirements. The RouteRover app will provide the following features:

Example:
USE-1: The RouteRover app will enable users to get real-time notifications and alerts with a maximum of three interactions.
USE-2: The application will offer a straightforward interface for users to alert about incidents, with just two screen taps needed

2.4.3 Performance

Meeting the performance criteria of the RouteRover app is essential for ensuring user contentment. The system must achieve the performance benchmarks listed below.

Example:
PER-1: Users can expect to receive 95% of traffic and event updates within 3 seconds of being reported.
PER-2: The app must be able to handle a minimum of 500 simultaneous users without any decrease in performance.

2.4.4 Security

The RouteRover app needs to follow strict security measures to safeguard user data and privacy. The security needs encompass: Example:
SEC-1: The system must guarantee that user data is encrypted using AES-256 encryption while being transmitted and stored.
SEC-2: The app will use role-based access control to limit administrative tasks to authorized users.

2.5 Domain Model

Create a representation of the domain model for your project.

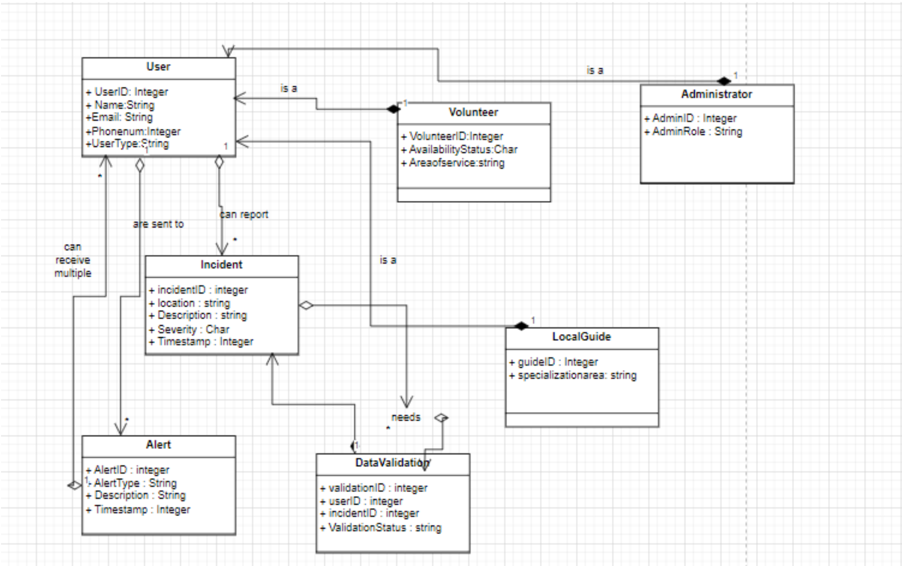


Figure 2.3: Domain Model for RouteRover.

Chapter 3

System Overview

RouteRover is a mobile app that provides real-time information for efficient and safe navigation through crowdsourcing. This app centers around delivering location based updates on traffic, crime alerts, events and safety warnings. Through the integration of AI, AR, this app aims to deliver a dynamic user experience.

3.1 Architectural Design

Develop a modular program structure and explain the relationships between the modules to achieve the complete functionality of the system. This is a high-level overview of how the system's modules collaborate with each other in order to achieve the desired functionality.

Don't go into too much detail about the individual subsystems. The main purpose is to gain a general understanding of how and why the system was decomposed, and how the individual parts work together.

Provide a diagram showing the major subsystems and their connections.

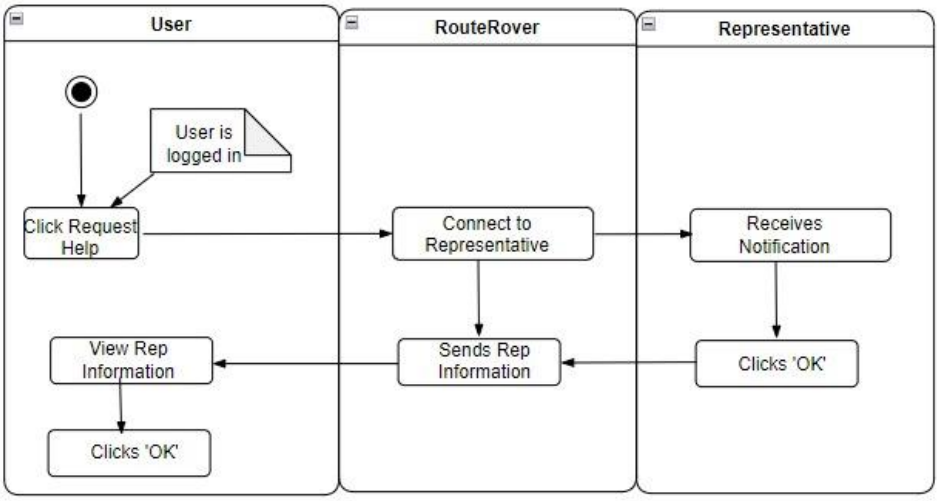
• In initial design stage create Box and Line Diagram for simpler representation of the systems • After finalizing architecture style/pattern diagram (MVC, Client-Server, Layered, Multi-tiered) create a detailed mapping modules/components to each part of the architecture

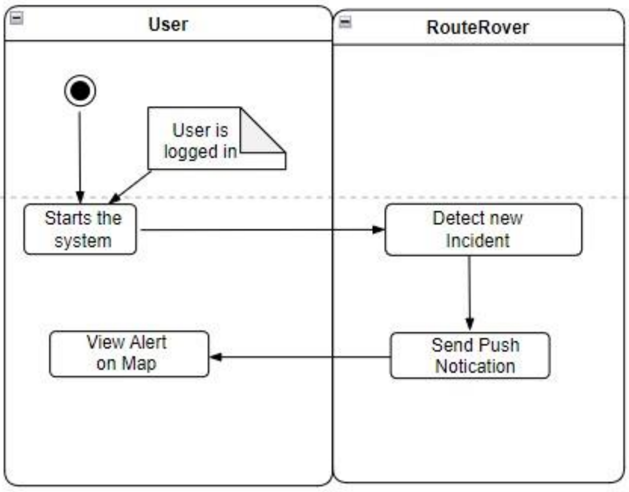
3.2 Design Models

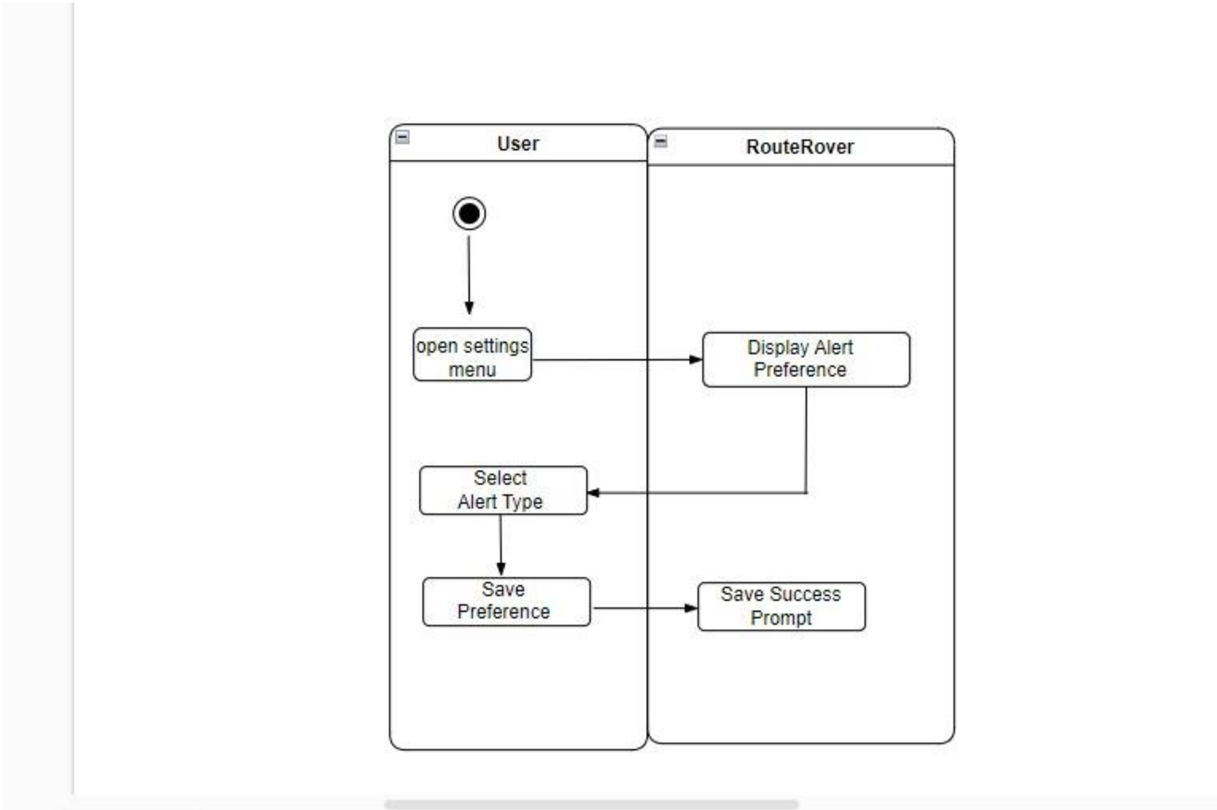
Create design models as are applicable to your system. Provide detailed descriptions with each of the models that you add. Also ensure visibility of all diagrams.

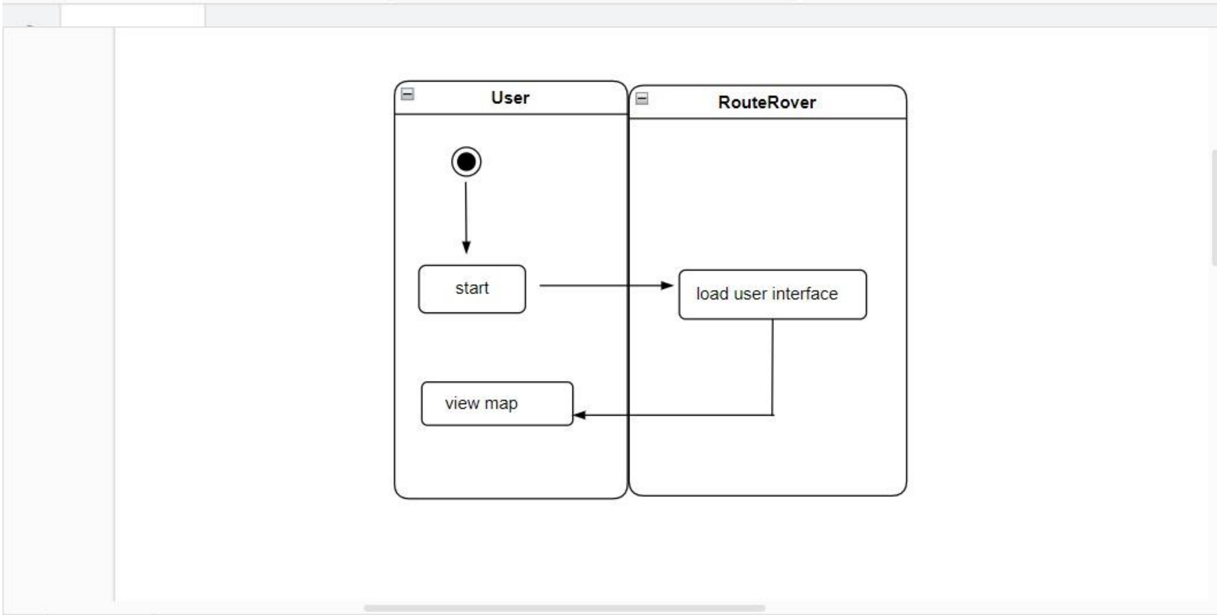
Design Models for Object Oriented Development Approach

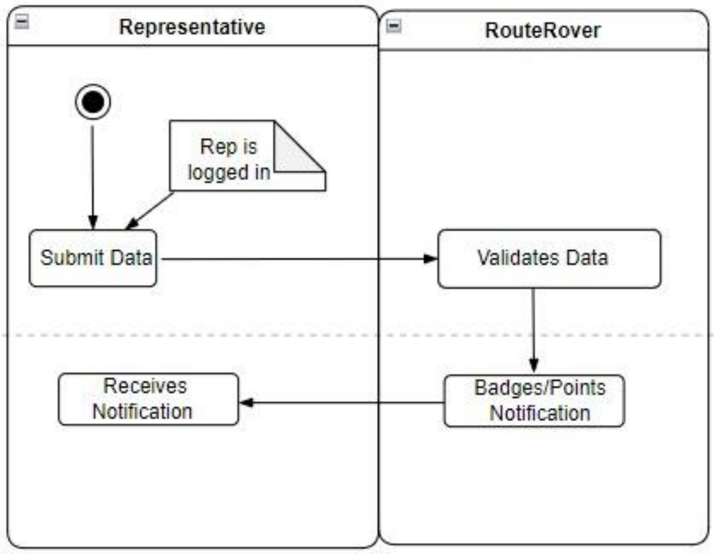
3.2.1 Activity Diagrams



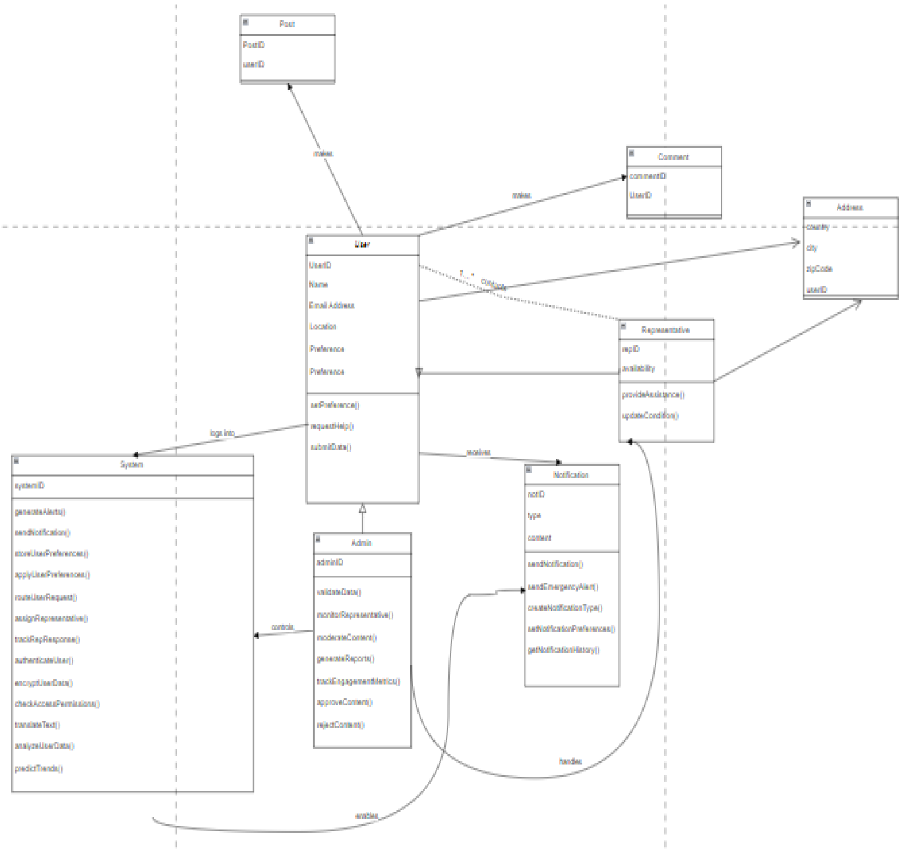



User
RouteRover
open settings menu
Display Alert Preference
Select Alert Type
Save Preference
Save Success Prompt

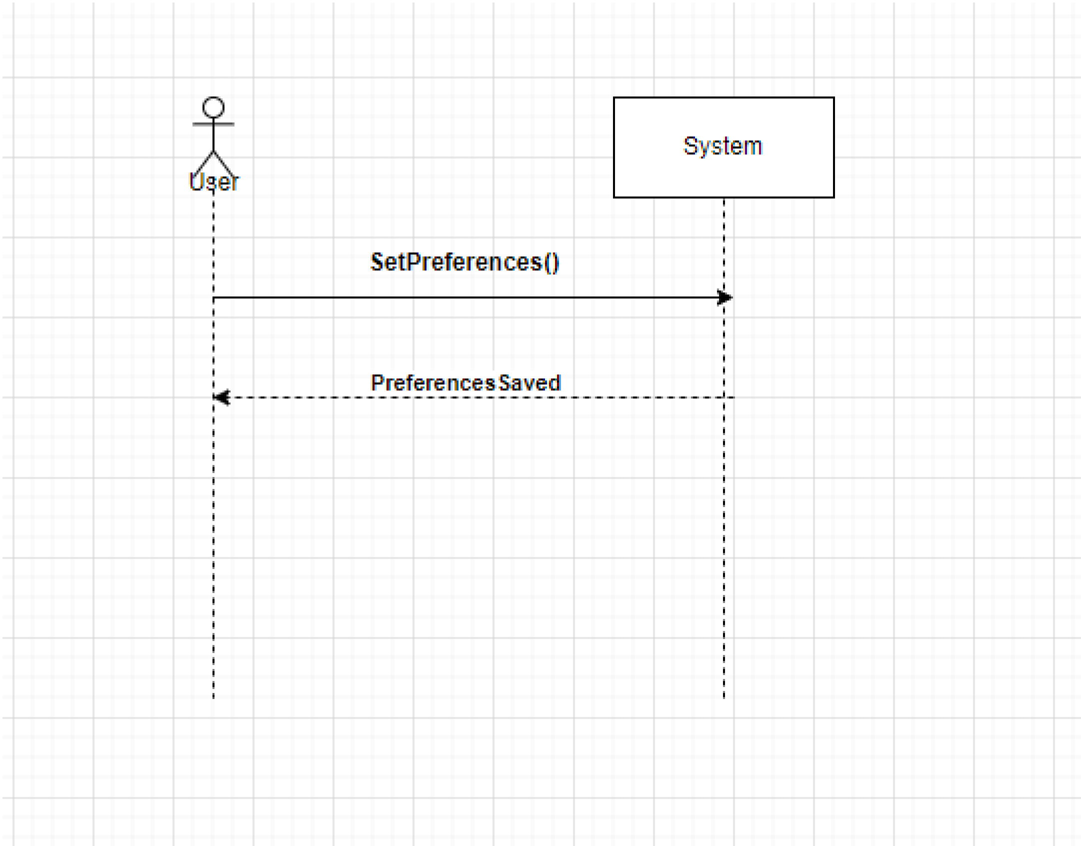

User
RouteRover
start
load user interface
view map



3.2.2 Class Diagram



3.2.3 System Sequence Diagram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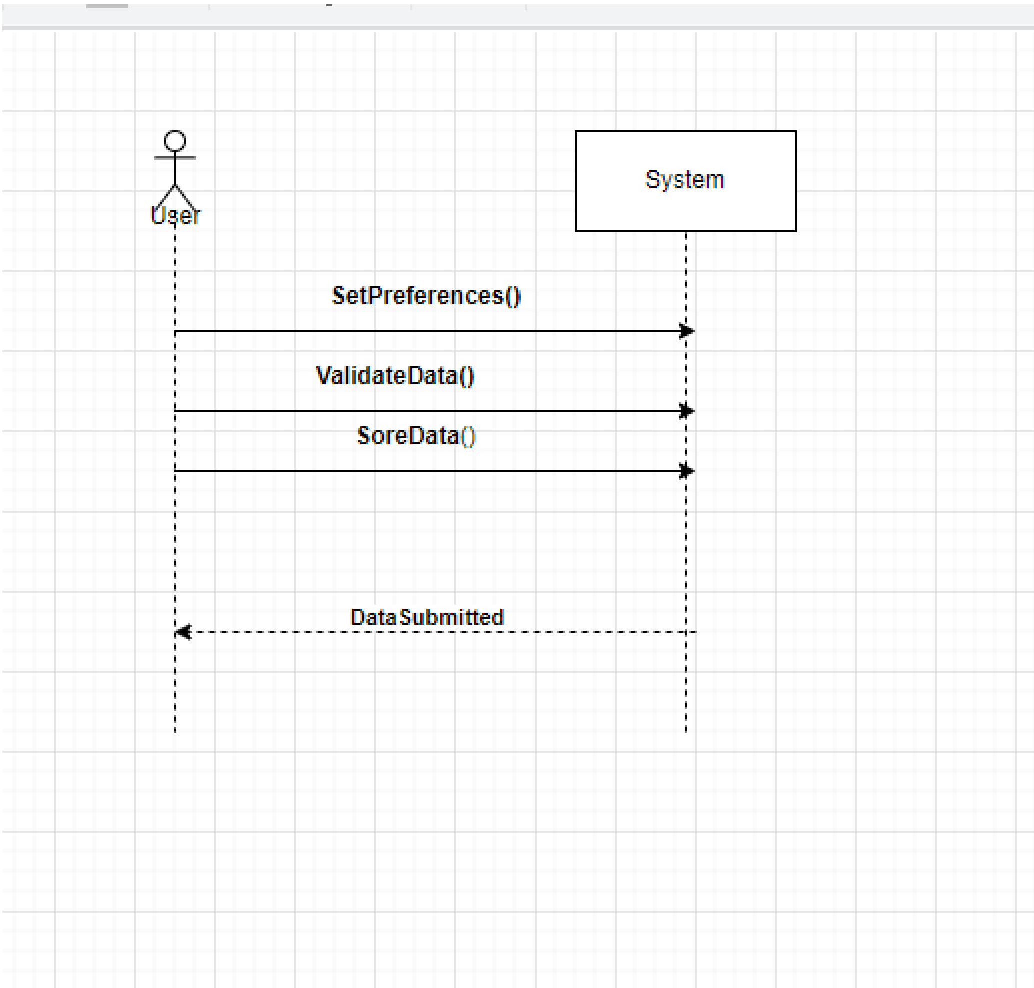
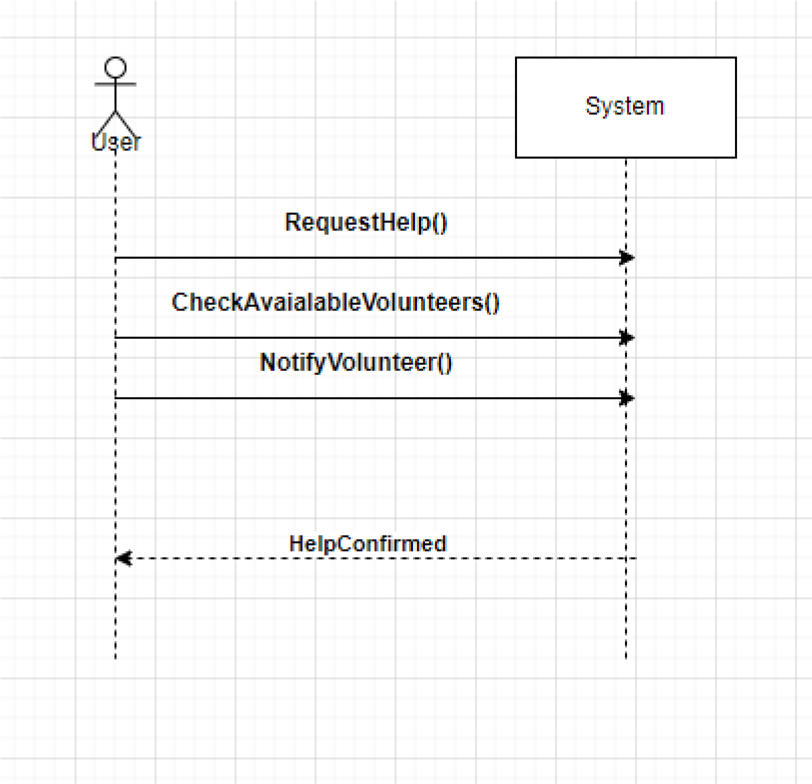
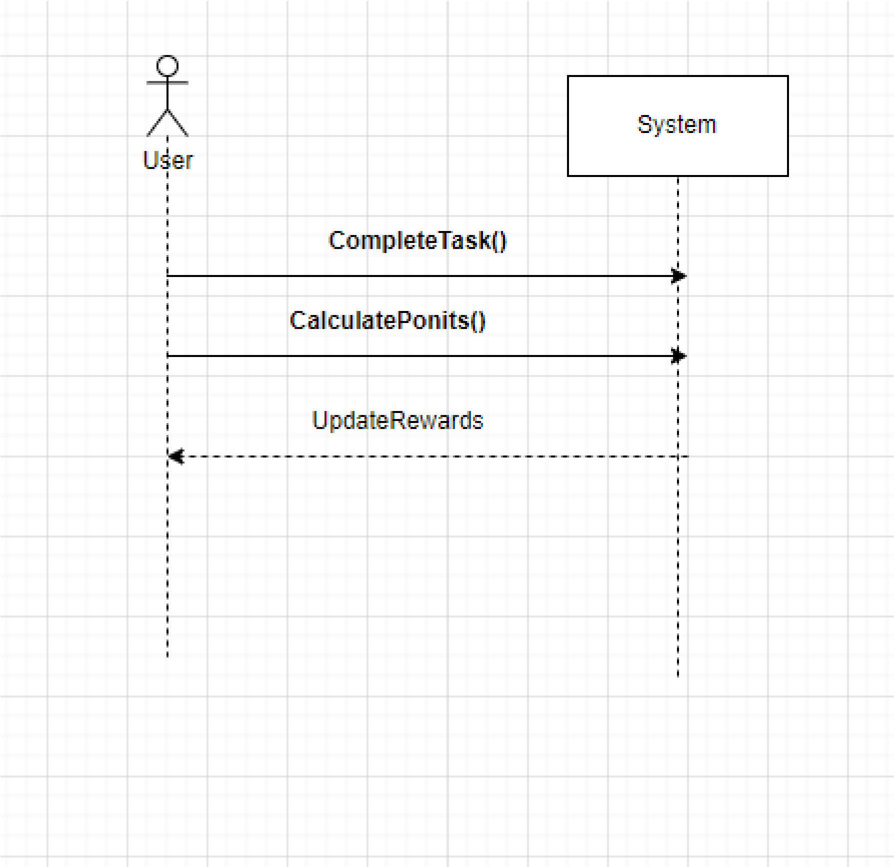
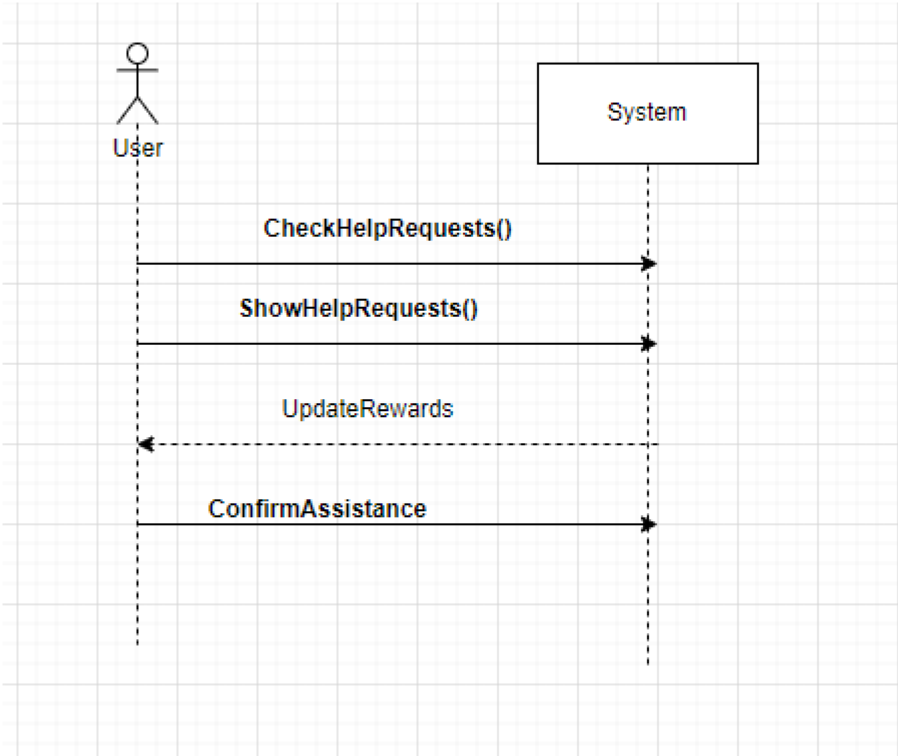
PreferencesSaved


System
User
SetPreferences()
ValidateData()
SoreData()
DataSubmitted


User
System
RequestHelp()
CheckAvaialableVolunteers()
NotifyVolunteer()
HelpConfirmed


User
System
CompleteTask()
CalculatePonits()
UpdateRewards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System
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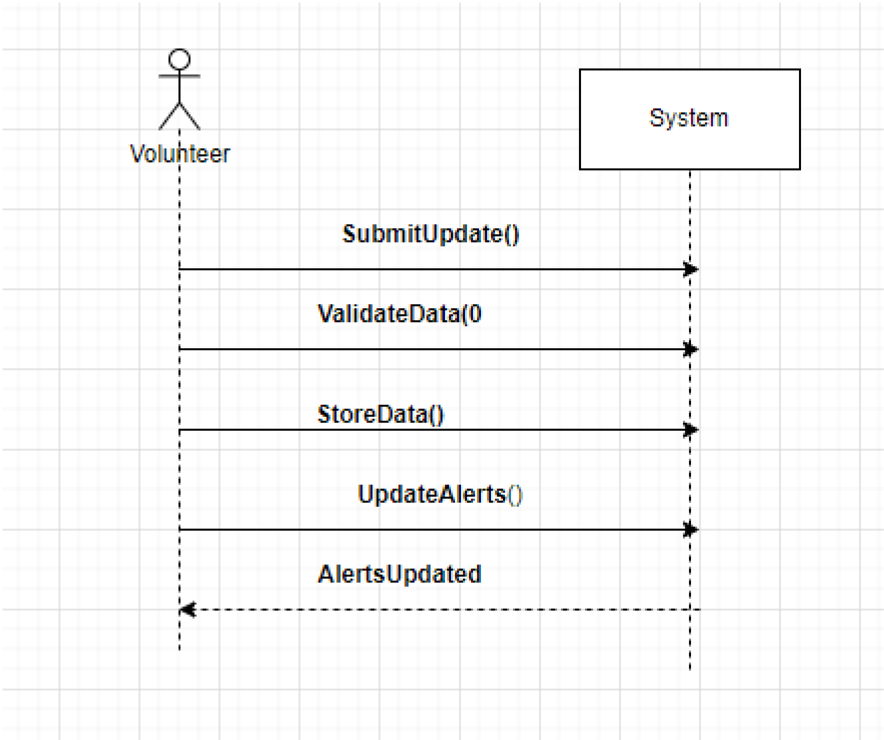
ShowHelpRequests()
UpdateRewards
ConfirmAssistance


Volunteer
System
SubmitUpdate()
ValidateData(0
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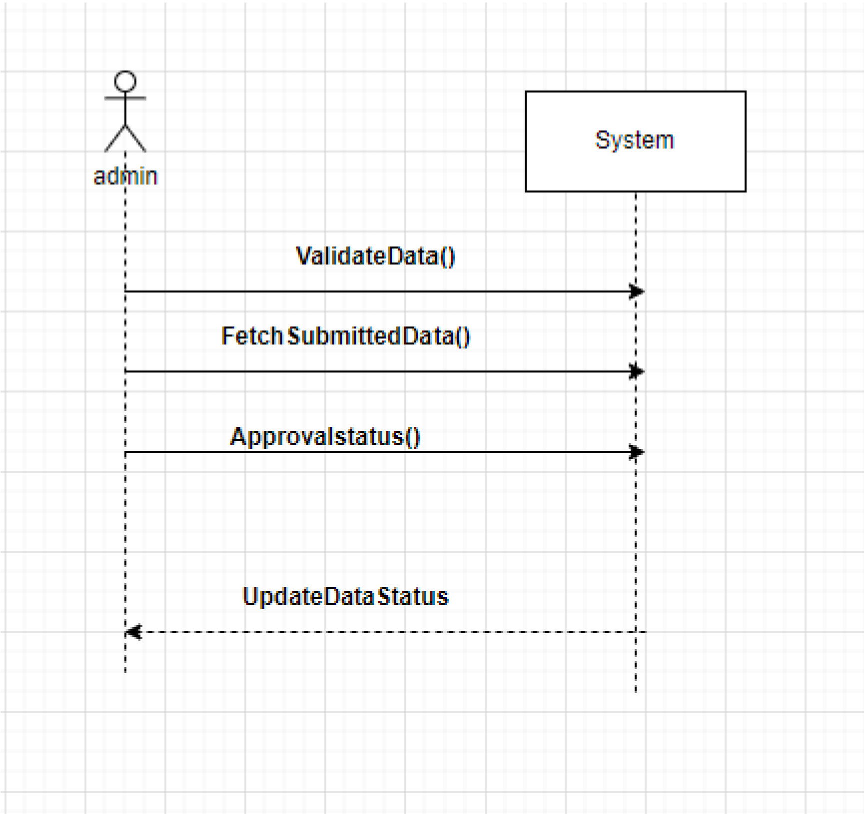
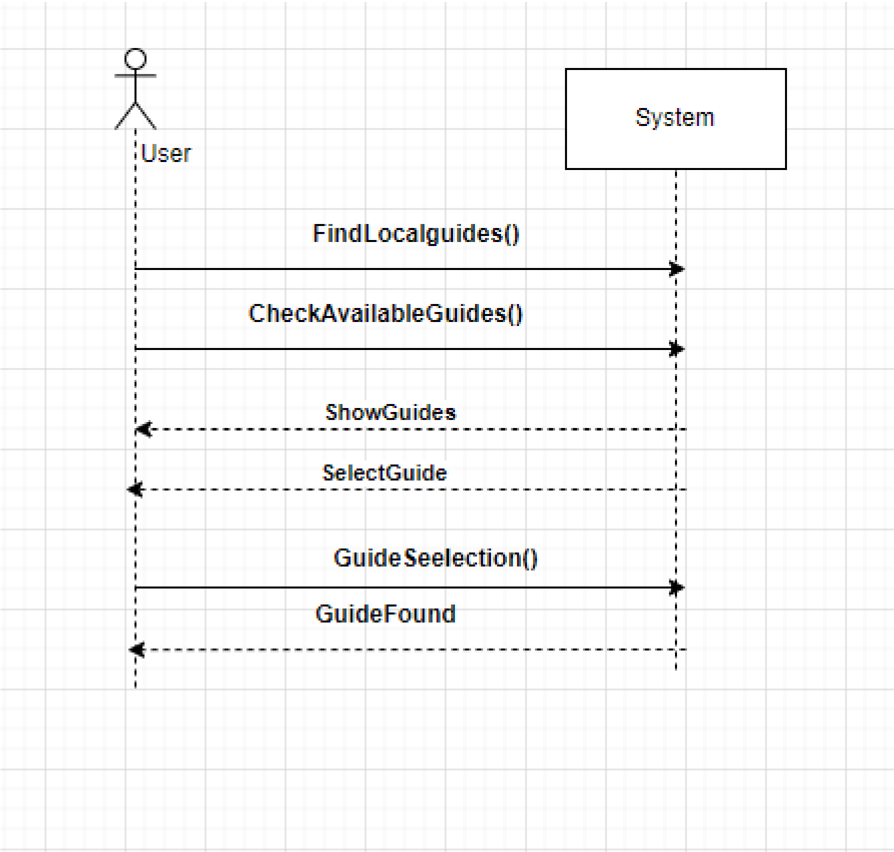
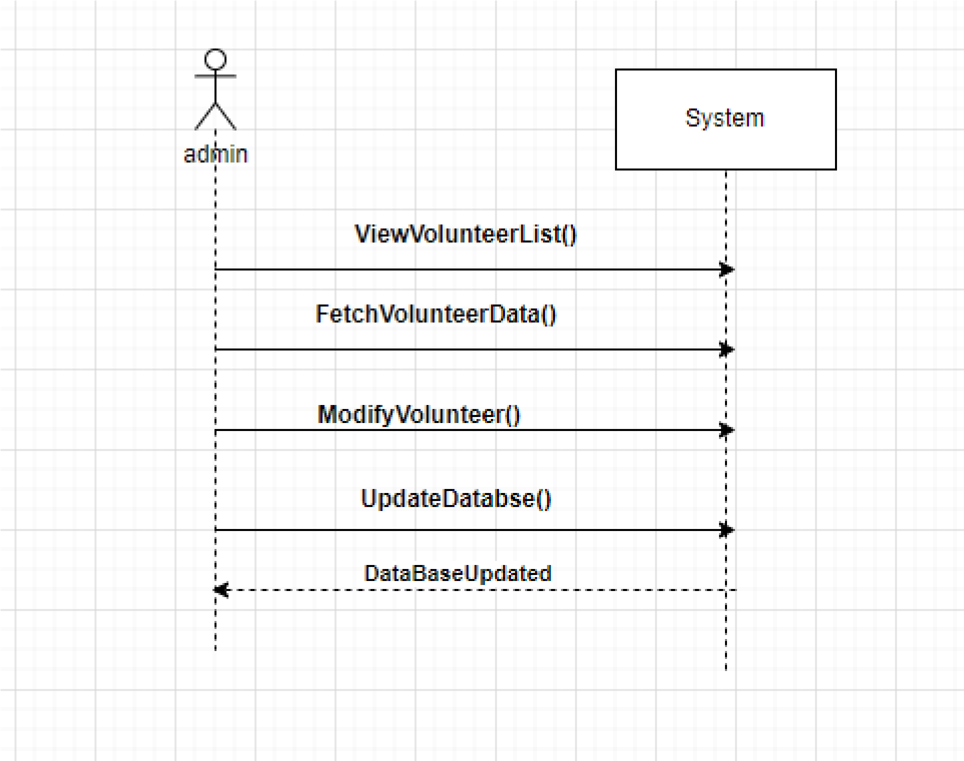
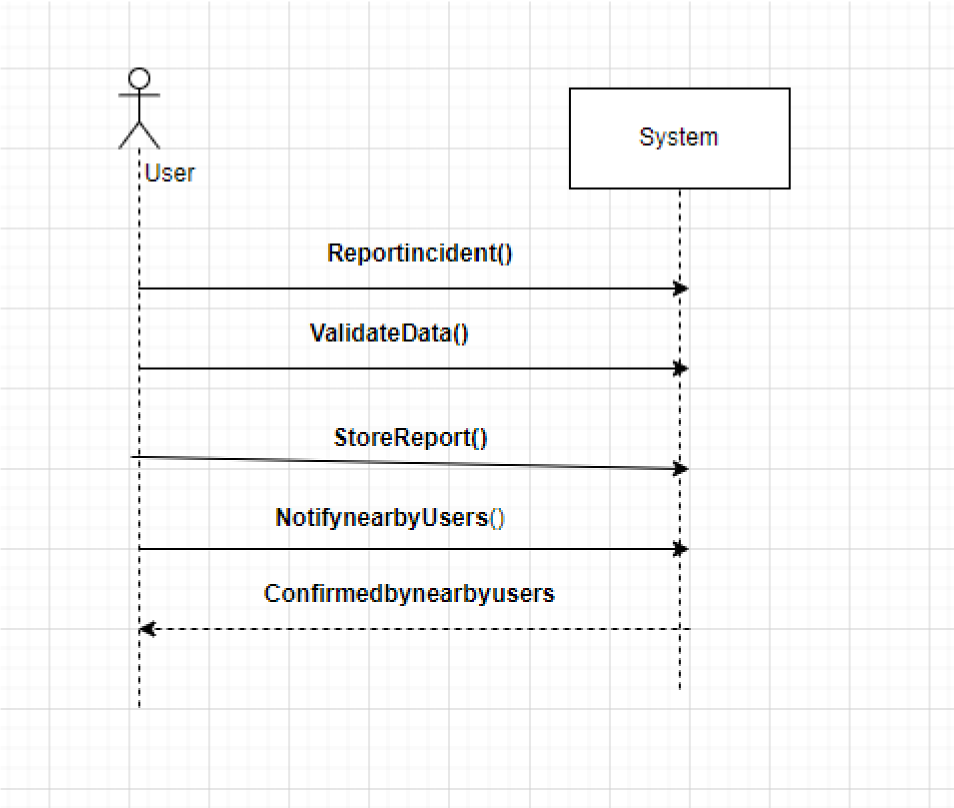
UpdateAlerts()
AlertsUpdated


admin
System
ValidateData()
Fetch SubmittedData()
Approvalstatus()
UpdateData Status


User
System
FindLocalguides()
CheckAvailableGuides()
ShowGuides
SelectGuide
Guide Seelection()
GuideFound


admin
System
ViewVolunteerList()
FetchVolunteerData()
ModifyVolunteer()
UpdateDatabse()
DataBaseUpdated


User
System
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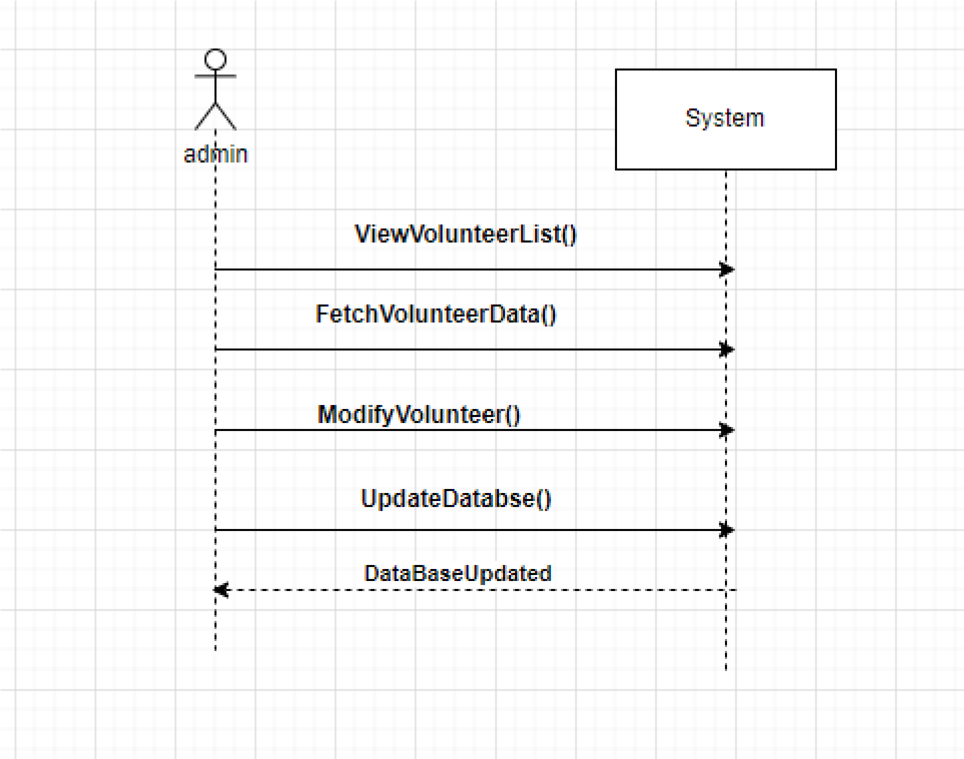
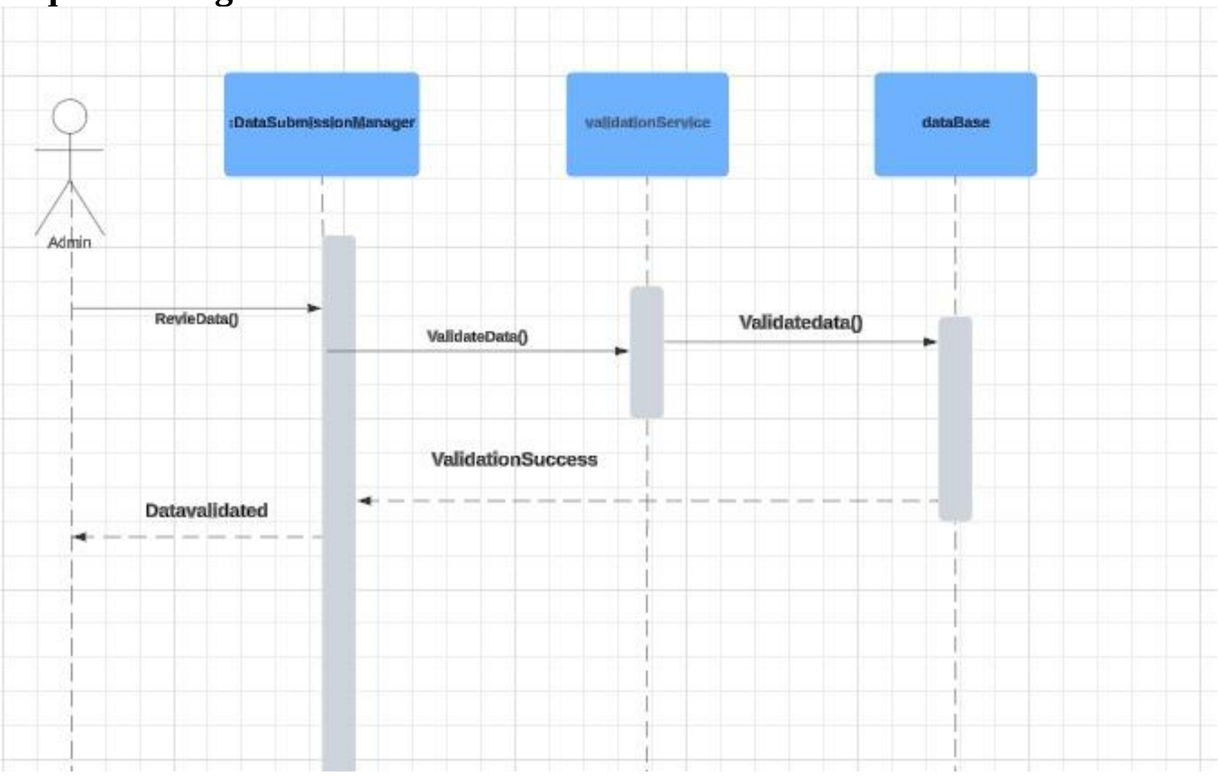
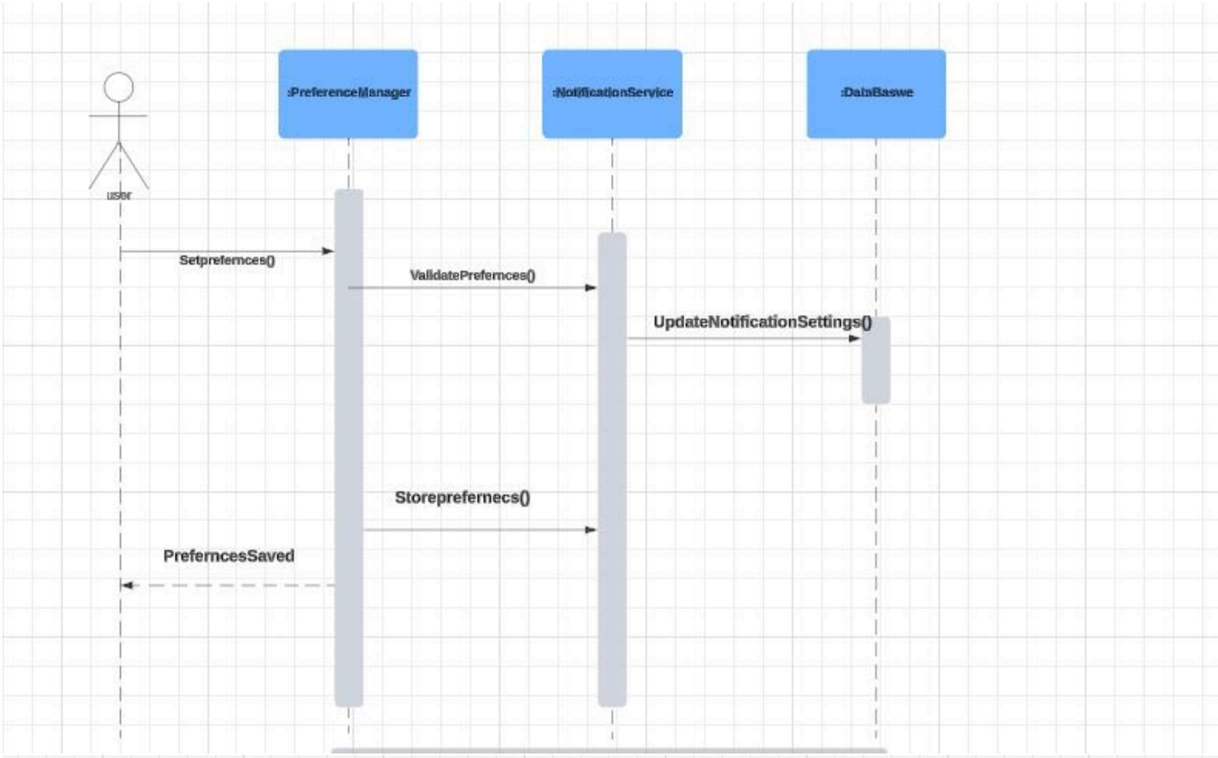
ValidateData()
StoreReport()
NotifynearbyUsers()
Confirmedbynearbyusers



3.2.4 Sequence Diagram

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:NotificationService
:DataBaswe
user
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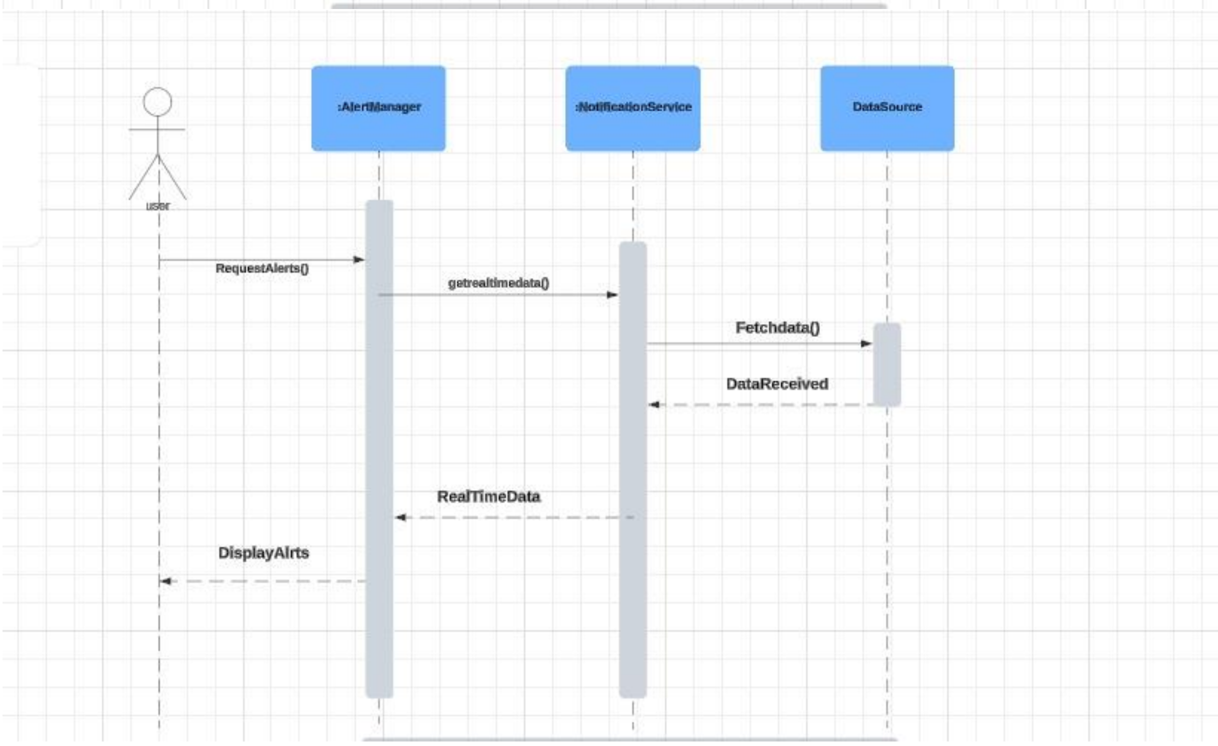
ValidatePrefernces()
UpdateNotificationSettings()
Storeprefernecs()
PreferncesSaved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:NotificationService
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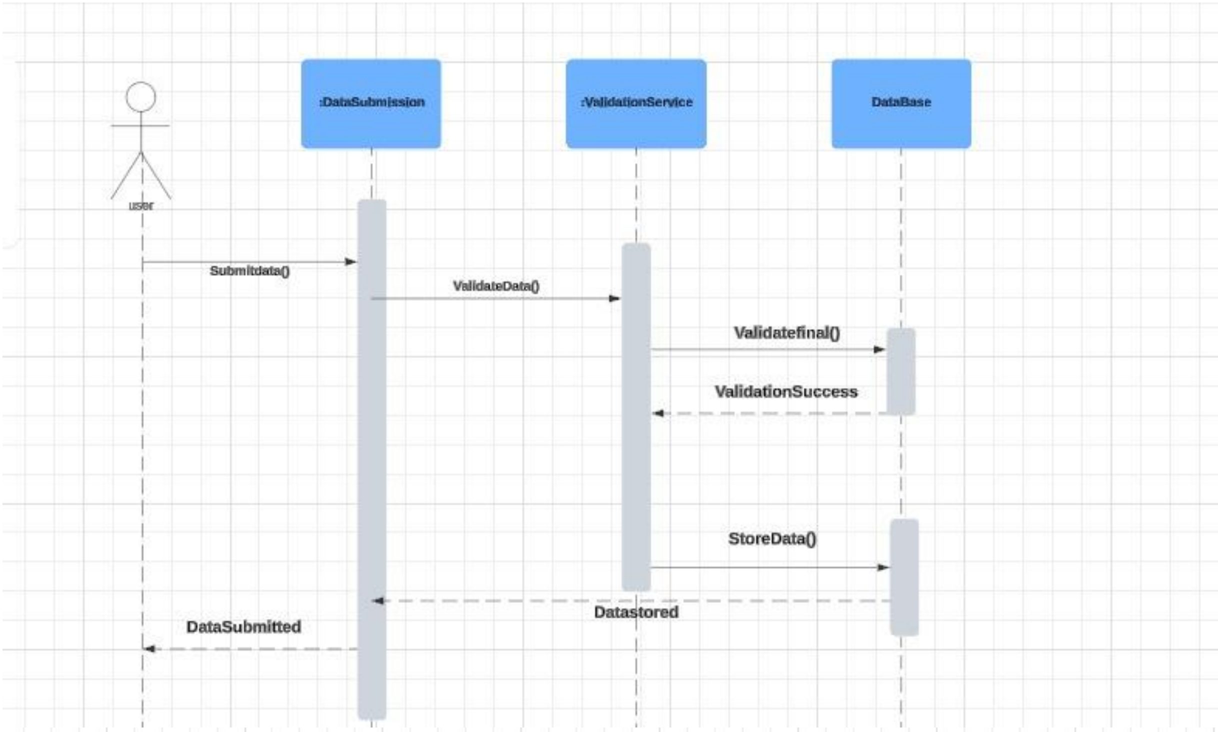
Fetchdata()
DataReceived
RealTimeData
DisplayAlrts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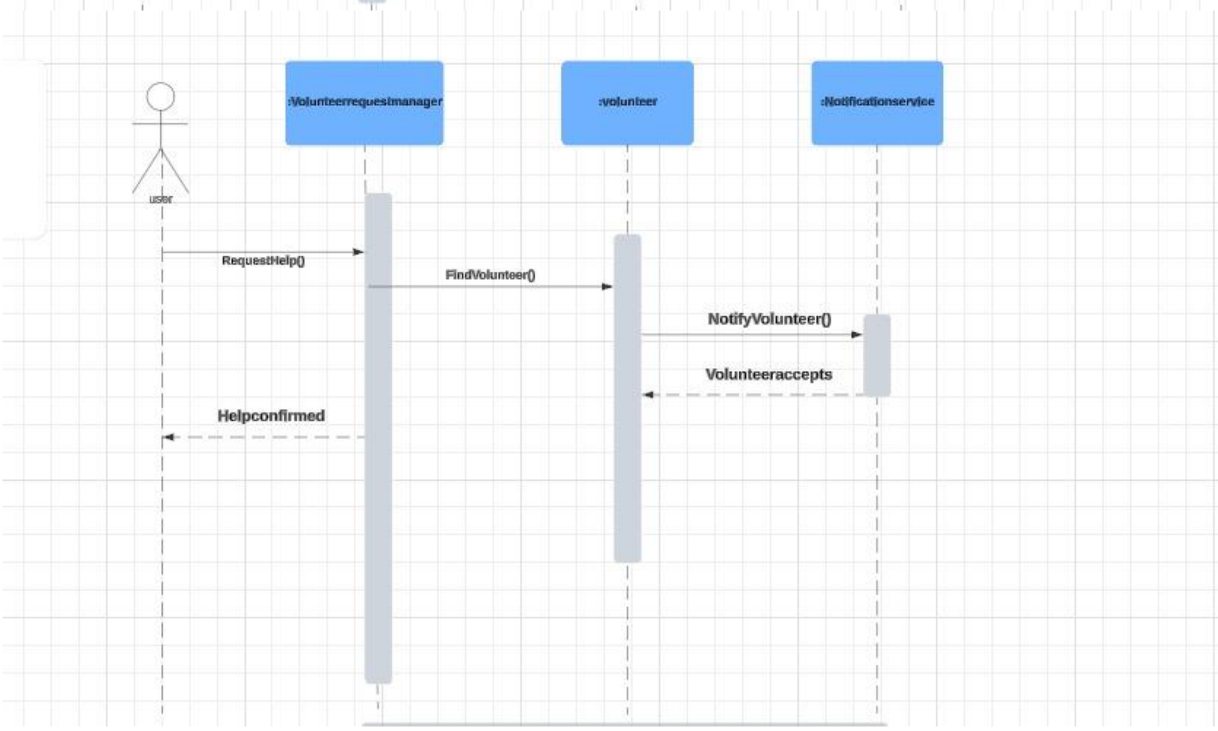
ValidationSuccess
StoreData()
Datastored
DataSubmitted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:volunteer
:Notificationservice
user
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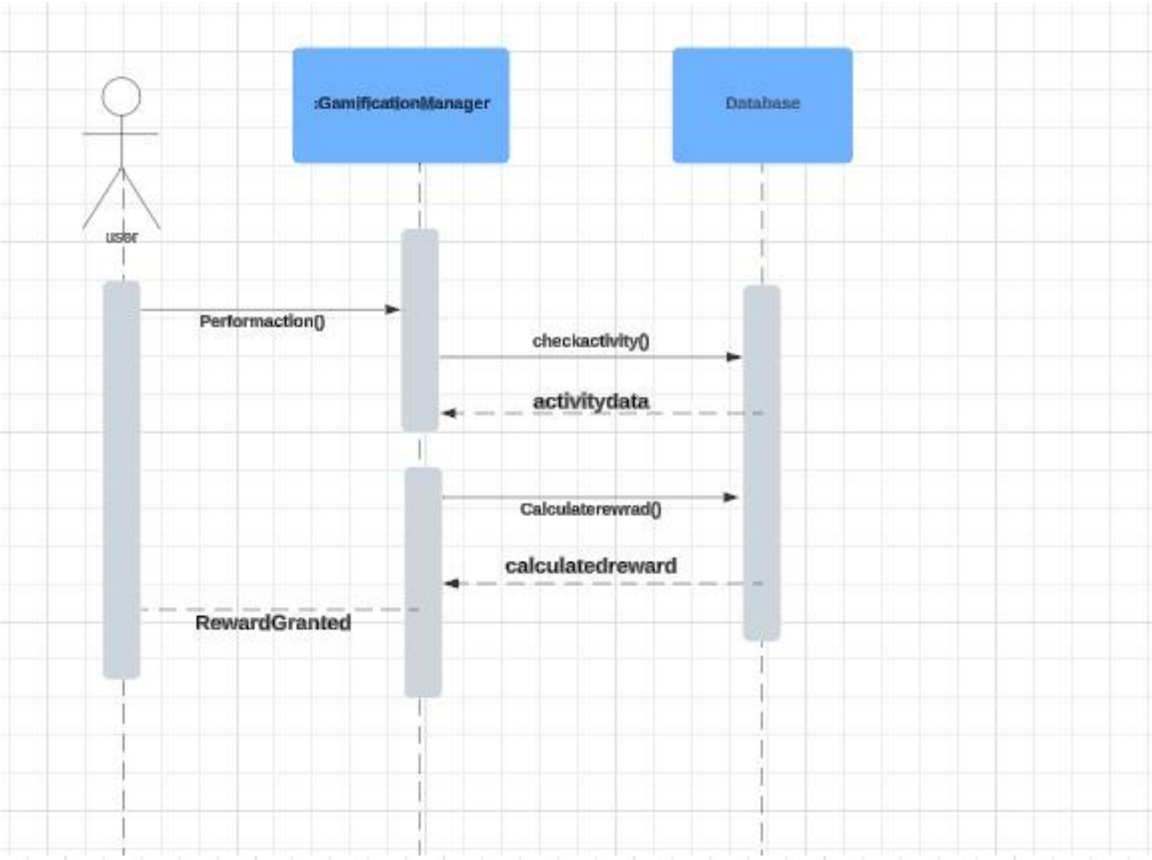
FindVolunteer()
NotifyVolunteer()
Volunteeraccepts
Helpconfirmed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Database
user
Performaction()
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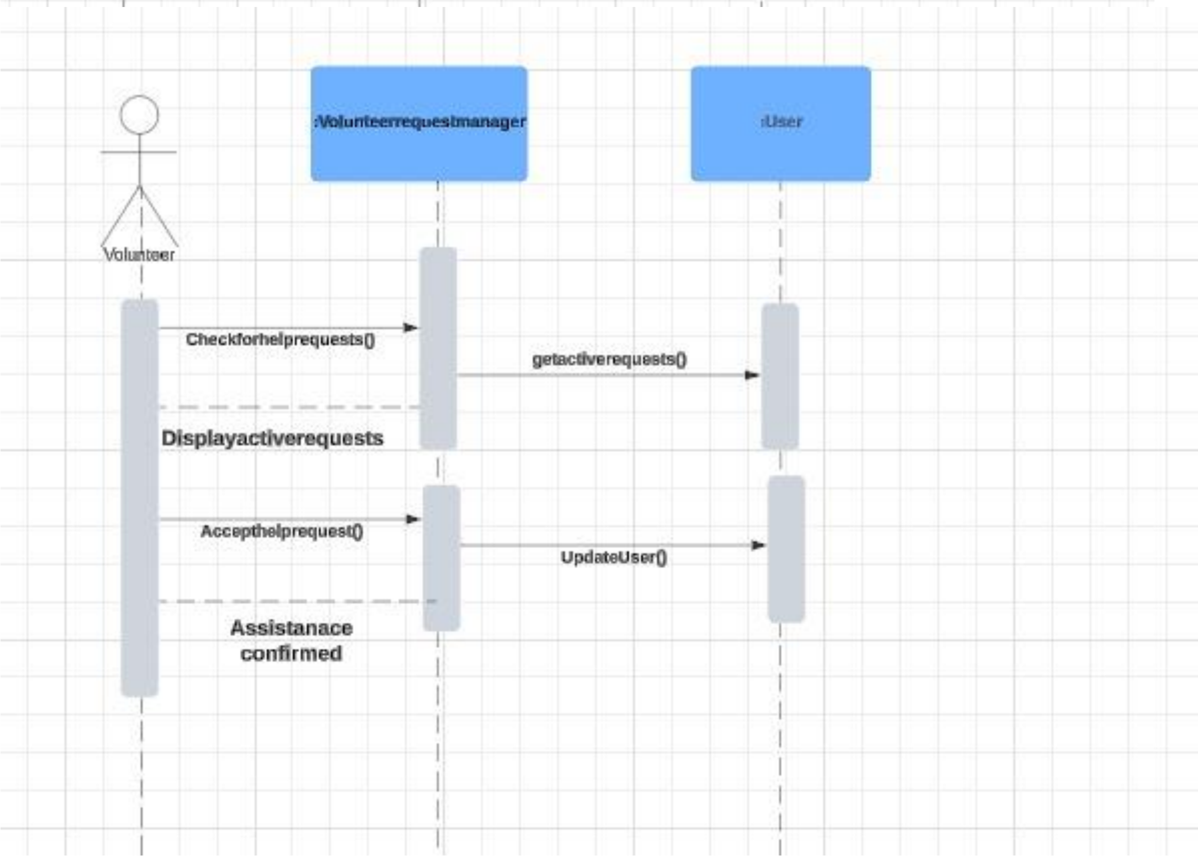
activitydata
Calculaterewrad()
calculatedreward
RewardGranted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:User
Volunteer
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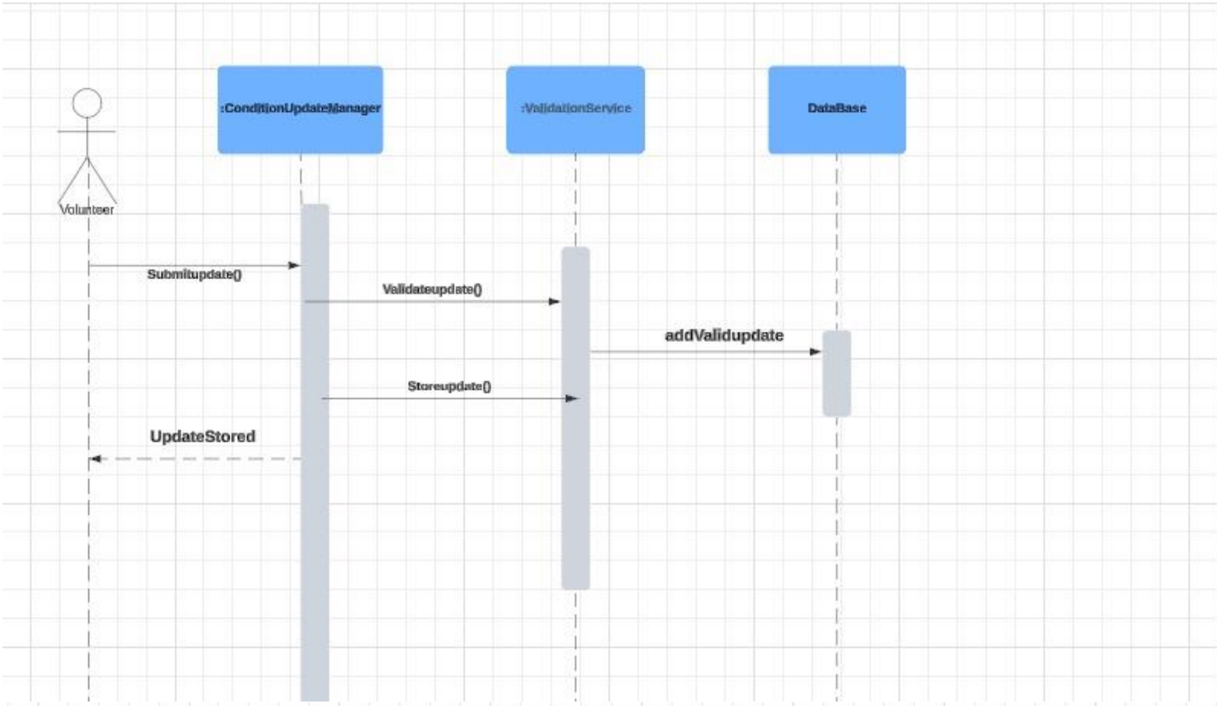
getactiverequests()
Displayactiverequests
Accepthelprequest()
UpdateUser()
Assistanace
confirmed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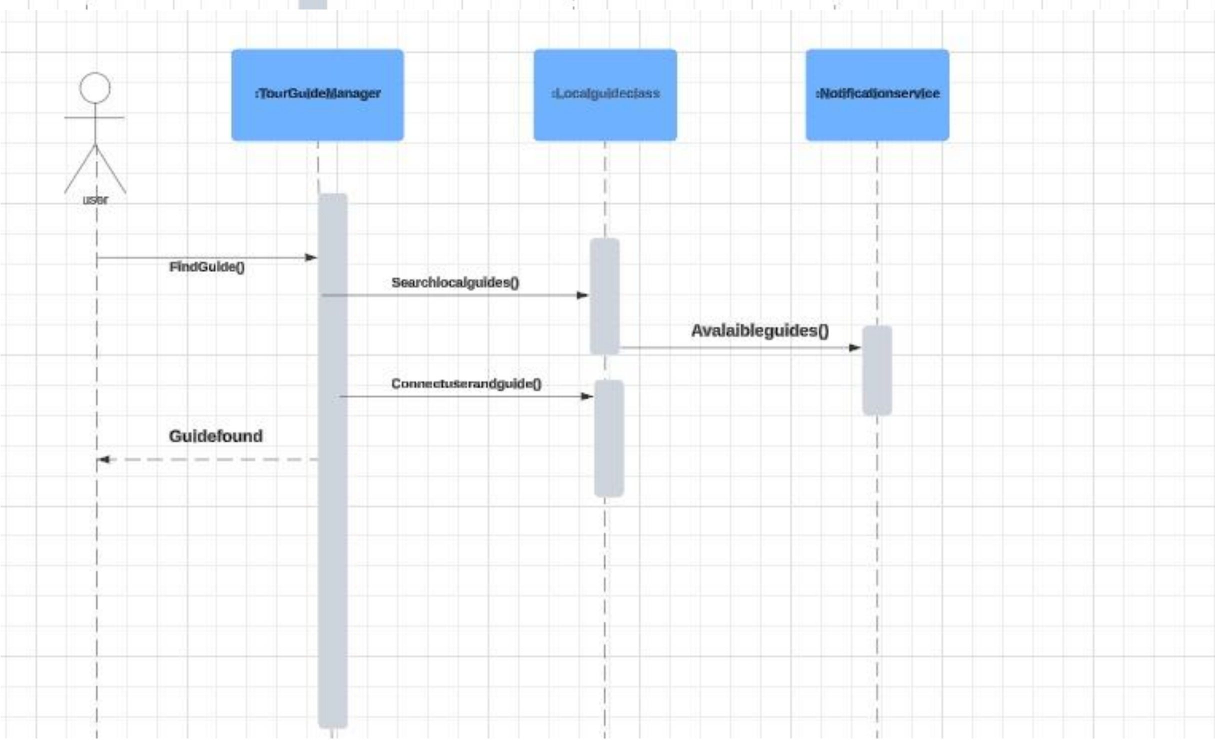
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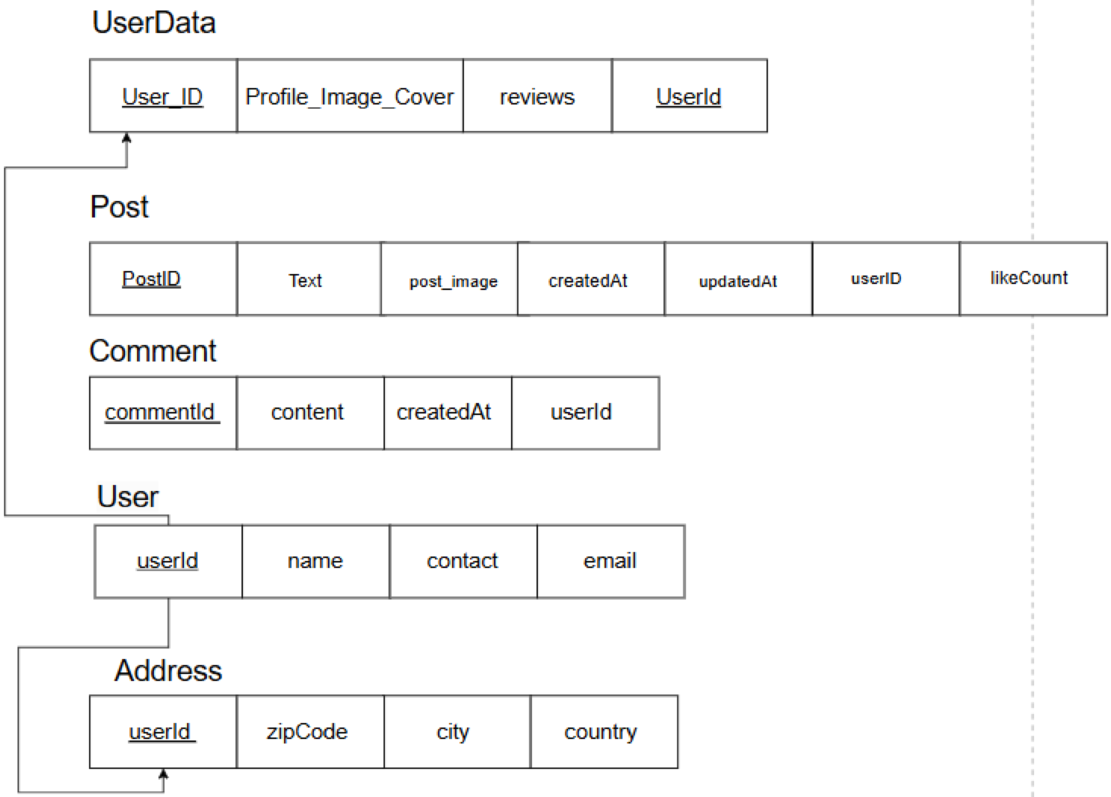
user
FindGuide()
Searchlocalguides()
Avalaibleguides()
Connectuserandguide()
Guidefound

3.2.5 ERD

ERD (Schema)



3.3 Data Design

Explain how the information domain of your system is transformed into data structures. Describe how the major data or system entities are stored, processed, and organized.

List any databases or data storage items.

Bibliography

[1] A Kolyshkin and S Nazarovs. Stability of slowly diverging flows in shallow water. *Mathematical Modeling and Analysis*, 2007.